Stock Market Data Analysis Report

1. Introduction

Financial markets generate large volumes of data daily, including prices of stocks, commodities, and cryptocurrencies. Analyzing this data helps in understanding market trends, price fluctuations, and investment patterns.

This project focuses on analyzing stock market data using **Python, Pandas, Matplotlib, and Seaborn** to extract insights from multiple financial assets such as **Bitcoin, Gold, Apple, Tesla, and Crude Oil**.

2. Objective

The main objectives of this project are:

- To analyze price trends of different financial assets
- To handle missing values in financial data
- To clean and preprocess the dataset
- To identify relationships between different assets
- To visualize trends using graphs

3. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

4. Dataset Overview

- Total Rows: 1243
- Total Columns: 39

The dataset includes prices and volumes of:

- Commodities → Natural Gas, Crude Oil, Copper, Gold, Silver
- Cryptocurrencies → Bitcoin, Ethereum
- Stock Market → Apple, Tesla, Microsoft, Amazon, Google, etc.

5. Data Cleaning and Preprocessing

5.1 Dropping Unnecessary Column

```
df.drop("Unnamed: 0", axis=1, inplace=True)
```

This column was just an index and not useful for analysis.

5.2 Handling Missing Values

Missing values were found in several columns:

- Platinum_Vol → 607 missing values
- Copper_Vol → 37 missing values
- Silver_Vol → 47 missing values
- Crude_oil_Vol → 23 missing values

Method Used: Forward Fill (ffill)

```
df.fillna(method='ffill', inplace=True)
```

Explanation:
Forward fill replaces missing values with the previous available value, which is suitable for time-series financial data.

5.3 Converting Date Column

```
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)
```

This ensures proper date format for time-based analysis.

5.4 Fixing Object Data Types (Important Step)

Some price columns were in **string format due to commas** (e.g., "43,194.70").

```
df["Bitcoin_Price"] = df["Bitcoin_Price"].str.replace(',', '').astype(float)
df["Gold_Price"] = df["Gold_Price"].str.replace(',', '').astype(float)
df["Berkshire_Price"] = df["Berkshire_Price"].str.replace(',', '').astype(float)
```

6. Exploratory Data Analysis (EDA)

EDA was performed to understand:

- Price trends over time
- Relationships between assets
- Distribution of values

7. Data Visualization

7.1 Bitcoin Price Trend

```
plt.figure(figsize=(10,5))
```

```
plt.plot(df["Date"], df["Bitcoin_Price"])
plt.title("Bitcoin Price Trend")
plt.xlabel("Date")
plt.ylabel("Price")
plt.show()
```

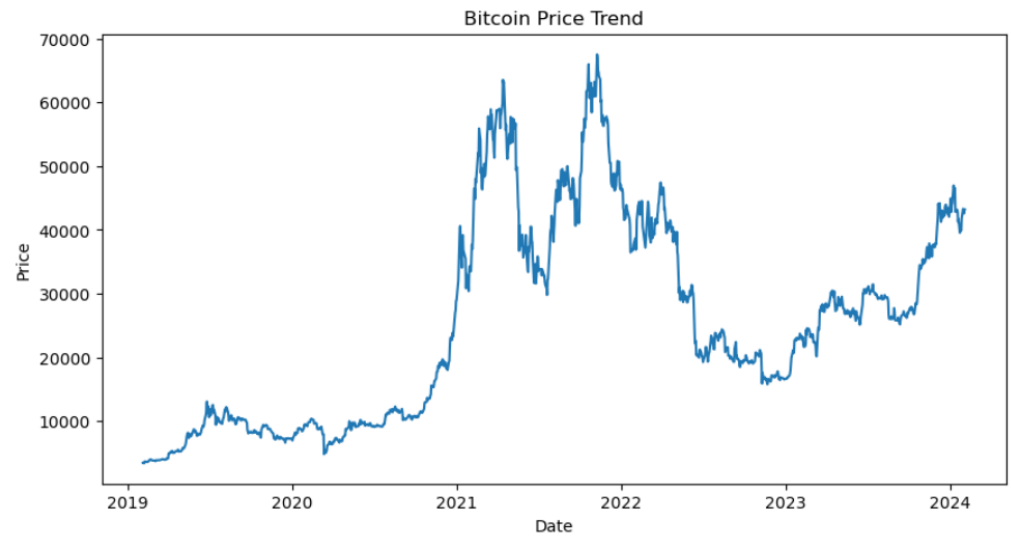

Figure 1: Bitcoin Price Trend

7.2 Gold Price Trend

```
plt.figure(figsize=(10,5))
plt.plot(df["Date"], df["Gold_Price"])
plt.title("Gold Price Trend")
plt.show()
```

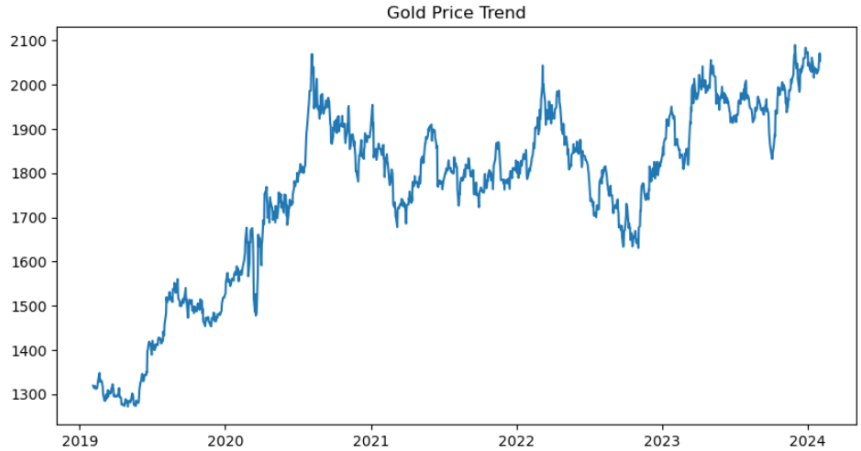

Figure 2: Gold Price Trend

7.3 Stock Comparison (Gold vs Bitcoin)

```
plt.figure(figsize=(10,5))

plt.plot(df["Date"], df["Gold_Price"], label="Gold")
plt.plot(df["Date"], df["Bitcoin_Price"], label="Bitcoin")

plt.title("Gold vs Bitcoin Price Trend")
plt.xlabel("Date")
plt.ylabel("Price")

plt.legend()

plt.show()
```

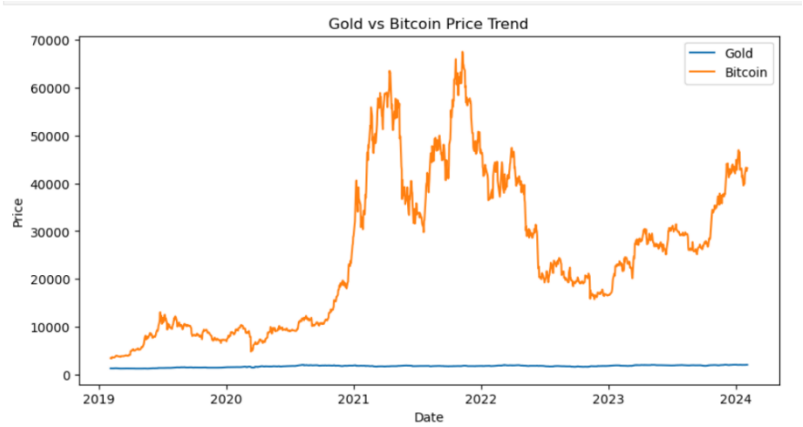

Figure 3: Stock Comparison

7.4 Correlation Heatmap

```
plt.figure(figsize=(12,8))

corr = df.corr(numeric_only=True)

sns.heatmap(corr, cmap="coolwarm")

plt.title("Market Correlation Heatmap")

plt.show()
```

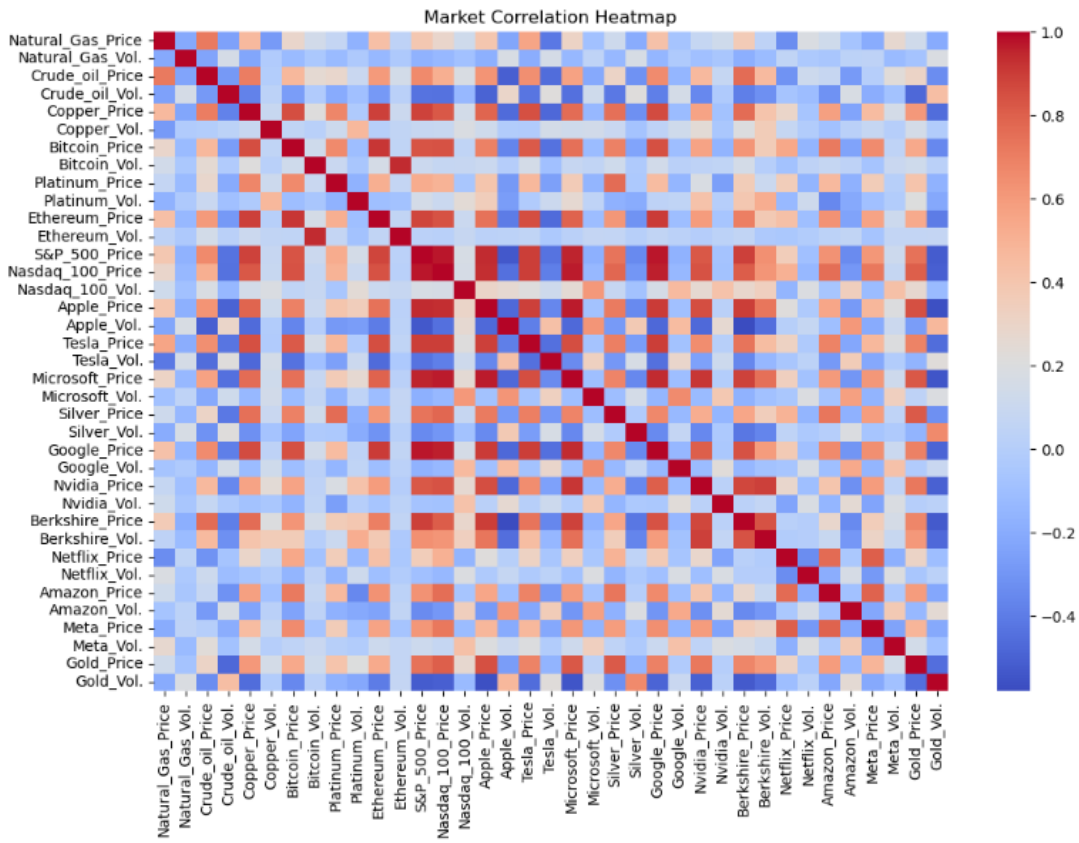

Figure 4: Correlation Heatmap

8. Key Insights

- Bitcoin shows high volatility compared to traditional assets
- Gold prices are relatively stable
- Stocks like Apple and Tesla show upward trends over time
- Some assets may show correlation, helping in portfolio analysis
- Missing data handling is crucial in financial datasets

9. Conclusion

This project successfully analyzed stock market data using Python. The dataset was cleaned, processed, and visualized to identify trends and relationships between financial assets.

Such analysis is useful for:

- Investors
- Financial analysts
- Market researchers

It helps in making **data-driven investment decisions**.

10. Future Improvements

- Use machine learning for price prediction
- Add more assets for deeper analysis
- Build a dashboard using Power BI
- Perform time-series forecasting